

VIVEK GOPALAKRISHNAN

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Research Overview

I am a 5th year PhD candidate in Medical Engineering and Medical Physics at the Harvard-MIT Program in Health Sciences and Technology. My research centers on *patient-specific machine learning*, developing personalized, physics-informed computer vision models capable of extracting 3D information from 2D imaging. I am particularly interested in applications in diagnostics, image-guided interventions, and surgical robotics.

Research Areas: *computer vision, medical physics, surgical robotics, interventional imaging, X-ray.*

Education

Massachusetts Institute of Technology
Ph.D. in Medical Engineering and Medical Physics

2021–Present
Advisor: Polina Golland

Johns Hopkins University
M.S.E. in Biomedical Engineering

2020–2021
Advisors: Joshua Vogelstein and Carey Priebe

Johns Hopkins University
B.S. in Biomedical Engineering

2017–2021
Advisor: Joshua Vogelstein

Research Experience

Computer Science and Artificial Intelligence Laboratory, MIT

2021–Present

- PhD candidate in the **Medical Vision Group** building intelligent intraoperative image guidance systems.
- Developed **DiffDRR**, an open-source differentiable X-ray renderer with over 50,000 downloads [W1].
- Developed **xvr**, the first 2D/3D registration algorithm to achieve consistent sub-mm accuracy [C1, P1].
- Building 3D vision systems for neurosurgery [J8, J9], orthopedics [J5, A1], and radiotherapy [C2].
- Collaborating with surgical robotics startups to integrate these algorithms into FDA-approved devices.

Department of Biomedical Engineering, Johns Hopkins University

2018–2021

- Undergraduate researcher in the **NeuroData Lab** studying statistical graph theory for neuroscience.
- Developed novel machine learning algorithms to analyze populations of graph-valued data [J4, J10].
- Applied these methods to discover neuroconnectively similar subtypes of autism spectrum disorder from multi-subject connectomics data [J3].

Fellowships and Awards

Rising Star in AI , University of Michigan	2025
MIT HEALS Graduate Fellowship (<i>9 months of full funding</i>), MIT	2025
Best Poster Award , MIT-MGB AI Cures Conference	2024
Neuroimaging Training Program Grant (<i>12 months of full funding</i>), MIT	2024
President’s Innovation Award , Society of Biomolecular Imaging and Informatics	2023
Takeda Fellowship (<i>9 months of full funding</i>), MIT	2023
Neuroimaging Training Program Grant (<i>12 months of full funding</i>), MIT	2022
Joseph C. Pistrutto Research Fellowship , Johns Hopkins University	2019
Provost’s Undergraduate Research Award , Johns Hopkins University	2020

Publications

Conferences

- C2 **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. “PolyPose: Deformable 2D/3D Registration via Polyrigid Transforms”. *Neural Information Processing Systems (NeurIPS)*, 2025.
- C1 **Vivek Gopalakrishnan**, Neel Dey, and Polina Golland. “Intraoperative 2D/3D Image Registration via Differentiable X-ray Rendering”. *Computer Vision and Pattern Recognition (CVPR)*, 2024.
🏆 **Best Poster Award** at the MIT-MGB AI Cures Conference (2024).

Journals

- J10 **Vivek Gopalakrishnan**, Jaewon Chung, Eric Bridgeford, Benjamin Pedigo, Jesús Arroyo, Lucy Upchurch, G. Allan Johnson, Nian Wang, Youngser Park, Carey Priebe, and Joshua Vogelstein. “Multiscale Comparative Connectomics”. *Imaging Neuroscience*, 2025.
- J9 Sarah Frisken, **Vivek Gopalakrishnan**, David Chlorogiannis, Nazim Haouchine, Alexandre Cafaro, Alexandra Golby, William Wells, and Rose Du. “Spatiotemporally Constrained 3D Reconstruction from Biplanar Digital Subtraction Angiography”. *International Journal of Computer Assisted Radiography and Surgery*, 2025.
- J8 Charles Downs, Matthijs van der Sluijs, Sandra A.P. Cornelissen, Frank te Nijenhuis, Robert van Oostenbrugge, Wim H. van Zwam, **Vivek Gopalakrishnan**, Xucong Zhang, Ruisheng Su, and Theo van Walsum. “Improving Automatic Cerebral 3D-2D CTA-DSA Registration”. *International Journal of Computer Assisted Radiology and Surgery*, 2025.
- J7 Joshua Marchant, Natalie Ferris, Diana Grass, Magdalena Allen, **Vivek Gopalakrishnan**, Mark Olchanyi, ..., Hui Wang, Anastasia Yendiki, Susie Huang, Bruce Rosen, Randy Gollub. “Mesoscale Brain Mapping: Bridging Scales and Modalities in Neuroimaging”. *Neuroinformatics*, 2024.
- J6 Sarah Frisken, Nazim Haouchine, David Chlorogiannis, **Vivek Gopalakrishnan**, Alexandre Cafaro, William Wells, Alexandra Golby, Rose Du. “VESCL: An Open-Source 2D Vessel Contouring Library”. *International Journal of Computer Assisted Radiography and Surgery*, 2024.
- J5 Andrew Abumoussa, **Vivek Gopalakrishnan**, Benjamin Succop, Michael Galgano, Sivakumar Jaikumar, Yueh Lee, and Deb Bhowmick. “Machine Learning for Automated and Real-Time 2D-3D Registration of the Spine Using Only a Single Radiograph”. *Neurosurgical Focus*, 2023.
- J4 Jaewon Chung, Eric Bridgeford, Jesus Arroyo, Benjamin Pedigo, Ali Saad-Eldin, **Vivek Gopalakrishnan**, Liang Xiang, Carey Priebe, and Joshua Vogelstein. “Statistical Connectomics”. *Annual Review of Statistics and Its Application*, 2021.
- J3 Nian Wang, Robert Anderson, David Ashbrook, **Vivek Gopalakrishnan**, Youngser Park, Carey Priebe, Yi Qi, Rick Laoprasert, Joshua Vogelstein, Robert Williams, and G. Allan Johnson. “Variability and Heritability of Mouse Brain Structure: Microscopic MRI Atlases and Connectomes for Diverse Strains”. *NeuroImage*, 2020.
- J2 **Vivek Gopalakrishnan**, Eliezer Bose, Usha Nair, Yuwei Cheng, and Musie Ghebremichael. “Pre-HAART CD4+ T-Lymphocytes as Biomarkers of Post-HAART Immune Recovery in HIV-Infected Children with or without TB Co-Infection”. *BMC Infectious Diseases*, 2020.
- J1 Jong Soo Lee, Elijah Paintsil, **Vivek Gopalakrishnan**, and Musie Ghebremichael. “A Comparison of Machine Learning Techniques for Classification of HIV Patients with Antiretroviral Therapy-Induced Mitochondrial Toxicity”. *BMC Medical Research Methodology*, 2019.

Preprints

- P1 **Vivek Gopalakrishnan**, David Chlorogiannis, Andrew Abumoussa, Darren B. Orbach, Nazim Haouchine, Sarah Frisken, Neel Dey, and Polina Golland. “Rapid Patient-Specific Neural Networks for Intraoperative X-ray to Volume Registration”. *arXiv*, 2025 (Under Review).

Peer-Reviewed Workshops

- W5** Noe Bertramo, Gabriel Duguey, and **Vivek Gopalakrishnan**. “DiffUS: Differentiable Ultrasound Rendering from Volumetric Imaging”. *Advances in Simplifying Medical Ultrasound Workshop at MICCAI*, 2025.
- W4** Federica Facente, Benjamin Billot, **Vivek Gopalakrishnan**, Manasi Kattel, Wen Wei, Polina Golland, Hervé Delingette, Nicholas Ayache, and Pierre Berthet-Rayne. “Multi-stage CNN for Fast Registration of 3D Preoperative CTs to 2D Intraoperative X-rays”. *Collaborative Intelligence and Autonomy in Image-guided Surgery Workshop at MICCAI*, 2025.
- W3** Mohammadhossein Momeni*, **Vivek Gopalakrishnan***, Neel Dey, Polina Golland, and Sarah Frisken. “Differentiable Voxel-based X-ray Rendering Improves Sparse-View 3D CBCT Reconstruction”. *Machine Learning and the Physical Sciences Workshop at NeurIPS*, 2024.
- W2** **Vivek Gopalakrishnan**, Jingzhe Ma, and Zhiyong Xie. “Grad-CAMO: Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images”. *Computer Vision for Microscopy Image Analysis Workshop at CVPR*, 2024.
- 🏆 **President’s Innovation Award** from the Society of Biomolecular Imaging and Informatics.
- W1** **Vivek Gopalakrishnan** and Polina Golland. “Fast Auto-Differentiable Digitally Reconstructed Radiographs for Solving Inverse Problems in Intraoperative Imaging”. *Clinical Image-based Procedures Workshop at MICCAI*, 2022.

Abstracts

- A1** Michael Hachadorian and **Vivek Gopalakrishnan**. “Recovery of 3D Component Position in Reverse Shoulder Arthroplasty from Postoperative Radiographs via 2D/3D Registration”. *American Shoulder and Elbow Surgeons Fellows’ Symposium*, 2025.

Invited Talks

Recovering Latent 3D Structure and Dynamics from 2D Intraoperative X-ray Images

- Johns Hopkins University, Baltimore, MD 2026
- INRIA Centre at Université Côte d’Azur, Nice, France 2026
- Medical Physics Seminar Series, MGB/BIDMC/Dana-Farber, Boston, MA 2026

Patient-Specific Machine Learning for 3D Intraoperative Image Guidance

- University of Michigan | AI for Science Symposium, Ann Arbor, MI 2025
- United Imaging Intelligence, Boston, MA 2025
- Noah Medical, San Francisco, CA 2025
- Indian Institute of Science, Bangalore, India 2025
- GE HealthCare AI, Bangalore, India 2025

Differentiable X-ray Rendering for Fast Intraoperative 2D/3D Image Registration

- Boston Medical Imaging Workshop, MIT, Cambridge, MA 2024
- GE HealthCare AI, Bangalore, India 2024
- MIT Visual Computing Seminar, Cambridge, MA 2024

Learning Interpretable Single-Cell Morphological Profiles from 3D Cell Painting Images

- Computer Vision for Microscopy Image Analysis Workshop (CVPR 2024), Seattle, WA 2024
- Society of Biomolecular Imaging and Informatics, Boston, MA 2023

Fast Auto-Differentiable DRRs for Intraoperative Imaging Problems

- Boston Medical Imaging Workshop, Brigham and Women’s Hospital, Boston, MA 2022
- Image-Guided Neurosurgery Meeting, Boston, MA 2022
- Medical Image Computing and Computer Assisted Interventions CLIP, Singapore 2022

Multiscale Comparative Connectomics

- *Presentations by Undergraduates in Life Sciences and Engineering*, Baltimore, MD 2021
- *PathAI*, Online 2021
- *Neuromatch Conference*, Online 2020

Mentorship

Research

- Minsuk Choi (Masters), University of Michigan Fall 2025 & Spring 2026
Co-supervised with Dr. Jeong Joon Park, University of Michigan
Project: MRI-to-CT translation for radiation-free volumetric guidance in surgery and radiotherapy
- Zengtian Deng (PhD), UCLA Summer 2025
Co-supervised with Drs. Bibo Shi and Tao Zhao, Noah Medical
Project: Implementing 2D/3D registration [P1] in a deployed surgical robotic system
- Jonathan Tjandra (Undergrad), MIT Spring 2025–Spring 2026
Co-supervised with Professor Polina Golland, MIT
Project: Learning to identify lung nodules from chest radiographs from synthetic data
- Noe Bertramo (Masters) and Gabriel Duguey (Masters), MIT Summer 2025
Project: Developing a differentiable ultrasound rendering engine [W5]
- Michelle Wu (Undergrad), MIT Summer 2025 & Fall 2025
Co-supervised with Professor Polina Golland, MIT
Project: A fully automated system for postoperative evaluation of shoulder replacement surgery
- Erik Xie (Undergrad), MIT Fall 2024 & Spring 2025
Co-supervised with Professor Polina Golland, MIT
Project: Custom CUDA kernels for accelerated X-ray rendering
- Hossein Momeni (Undergrad), Truman State University Summer 2024 & Fall 2024
Co-supervised with Professor Sarah Frisken, Harvard Medical School
Project: Differentiable X-ray rendering improves Sparse-view 3D CBCT reconstruction [W3]
Current position: PhD student at UC Berkeley (EECS)

Community

- Graduate Resident Advisor, McCormick Hall, MIT 2024–2026

Teaching

Advances in Computer Vision (Graduate) Head Teaching Assistant, Massachusetts Institute of Technology	Spring 2025
NeuroData Design II (Undergraduate/Graduate) Head Teaching Assistant, Johns Hopkins University	Spring 2021
NeuroData Design I (Undergraduate/Graduate) Head Teaching Assistant, Johns Hopkins University	Fall 2020
Computational Cardiology Laboratory (Undergraduate) Teaching Assistant, Johns Hopkins University	Fall 2020
Linear Algebra (Undergraduate) Teaching Assistant, Johns Hopkins University	Fall 2020

Service

PhD Admissions Committee , Harvard-MIT Health Sciences and Technology	2025
Reviewer , IEEE Transactions Pattern Analysis and Machine Intelligence (TPAMI)	2025–2026
Reviewer , Medical Image Analysis (MedIA)	2025–2026
Reviewer , IEEE Transactions on Medical Imaging (🏆 <i>Distinguished Reviewer</i>)	2024–2026
Reviewer , Journal of Imaging Informatics in Medicine	2024–2025
Reviewer , Information Processing in Medical Imaging (IPMI)	2025
Reviewer , Medical Imaging Computing and Computer Assisted Interventions (MICCAI)	2024–2025
Reviewer , Medical Imaging with Deep Learning (MIDL)	2024

Software

xvr : Training patient-specific 2D/3D registration models in 5 min	2025
Primary author and maintainer (https://github.com/eigenvivek/xvr/)	
DiffDRR : GPU-accelerated & differentiable X-ray rendering in PyTorch	2022
Primary author and maintainer (https://github.com/eigenvivek/DiffDRR/)	

Work Experience

Consultant (Medical Imaging Machine Learning Scientist) , Noah Medical	2025–2026
Advising the implementation of 2D/3D registration for robotic bronchoscope navigation.	
Xellar Biosystems , Newton, MA	Summer 2023
Discovered mechanisms by which supervised deep learning models in drug discovery cheat by exploiting confounding factors, and developed an interpretability metric to quantify the level of confounding [W2].	